

Date:

**EXPERIMENT NO. 5****AIM: Understand and identify Layer-3 functionality.**

Tools required:

1. Desktop Computer
2. Cisco Packet Tracer

Simulate the different scenarios given below in the Cisco packet tracker and fill up the table.

**Note:** While applying an IP address, the student needs to allocate an IP address as per his/her student ID. For Example, if the student ID is 22cs005 then IP address allocation for the first network should start with 5.0.0.0. For subsequent networks, it should start with ID+1 i.e. 6.0.0.0, 7.0.0.0 and so on.

**Routing**

- A Router is a process of selecting a path along which the data can be transferred from the source to the destination. Routing is performed by a special device known as a router.
- A Router works at the network layer in the OSI model and the internet layer in the TCP/IP model
- A router is a networking device that forwards the packet based on the information available in the packet header and forwarding table.
- The routing algorithms are used for routing the packets. The routing algorithm is nothing but software responsible for deciding the optimal path through which a packet can be transmitted.
- The routing protocols use the metric to determine the best path for the packet delivery. The metric is the standard of measurement such as hop count, bandwidth, delay, current load on the path, etc. used by the routing algorithm to determine the optimal path to the destination.
- The routing algorithm initializes and maintains the routing table for the process of path determination.

**Types of routing**

- There are three types of routing as shown in Figure 5.1.

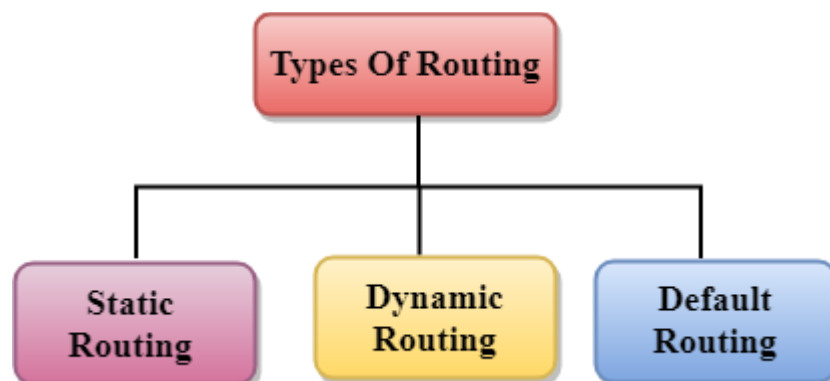


Figure 5.1

**Static routing:**

- Static Routing is also known as Non-Adaptive Routing.
- It is a technique in which the administrator manually adds the routes in a routing table.

**Dynamic routing:**

- It is also known as Adaptive Routing.
- It is a technique in which a router adds a new route in the routing table for each packet in response to the changes in the condition or topology of the network.
- Dynamic protocols are used to discover new routes to reach the destination.

**Default routing:**

- Default Routing is a technique in which a router is configured to send all the packets to the same hop device, and it doesn't matter whether it belongs to a particular network or not. A Packet is transmitted to the device for which it is configured in default routing.

**Routing Protocols**

Routing protocols can be either an interior protocol or an exterior protocol. An interior protocol handles intra-domain routing; an exterior protocol handles inter-domain routing.

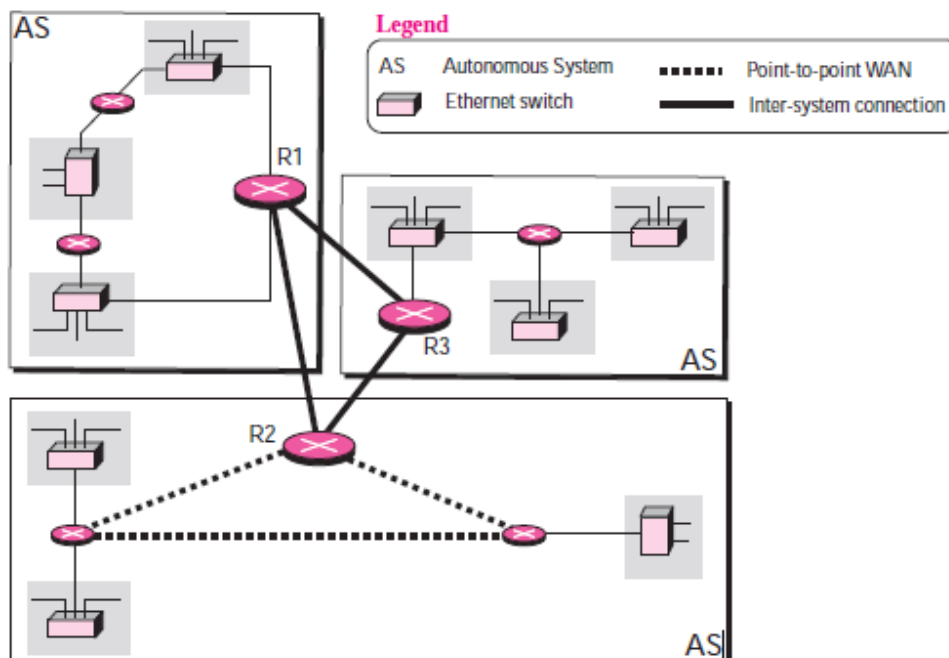


Figure 5.2 Intra-domain and Inter-domain routing protocol

Several intra-domain and inter-domain routing protocols are in use. We discuss two intra-domain routing protocols: distance vector and link state. We also introduce one inter-domain routing protocol: path vector.

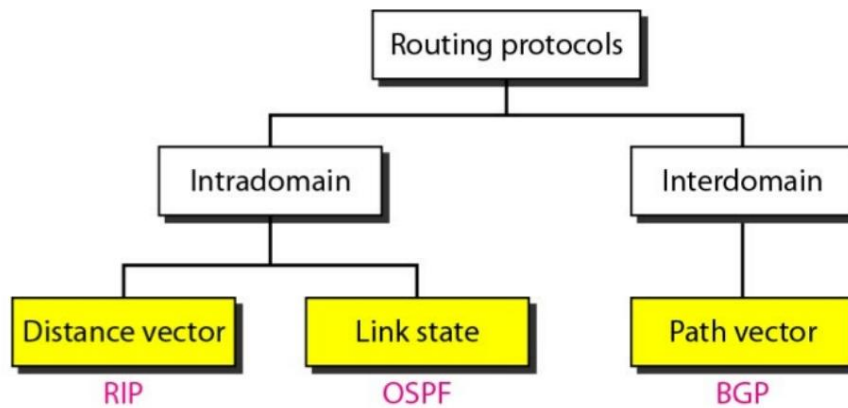


Figure 5.3 Routing Protocols

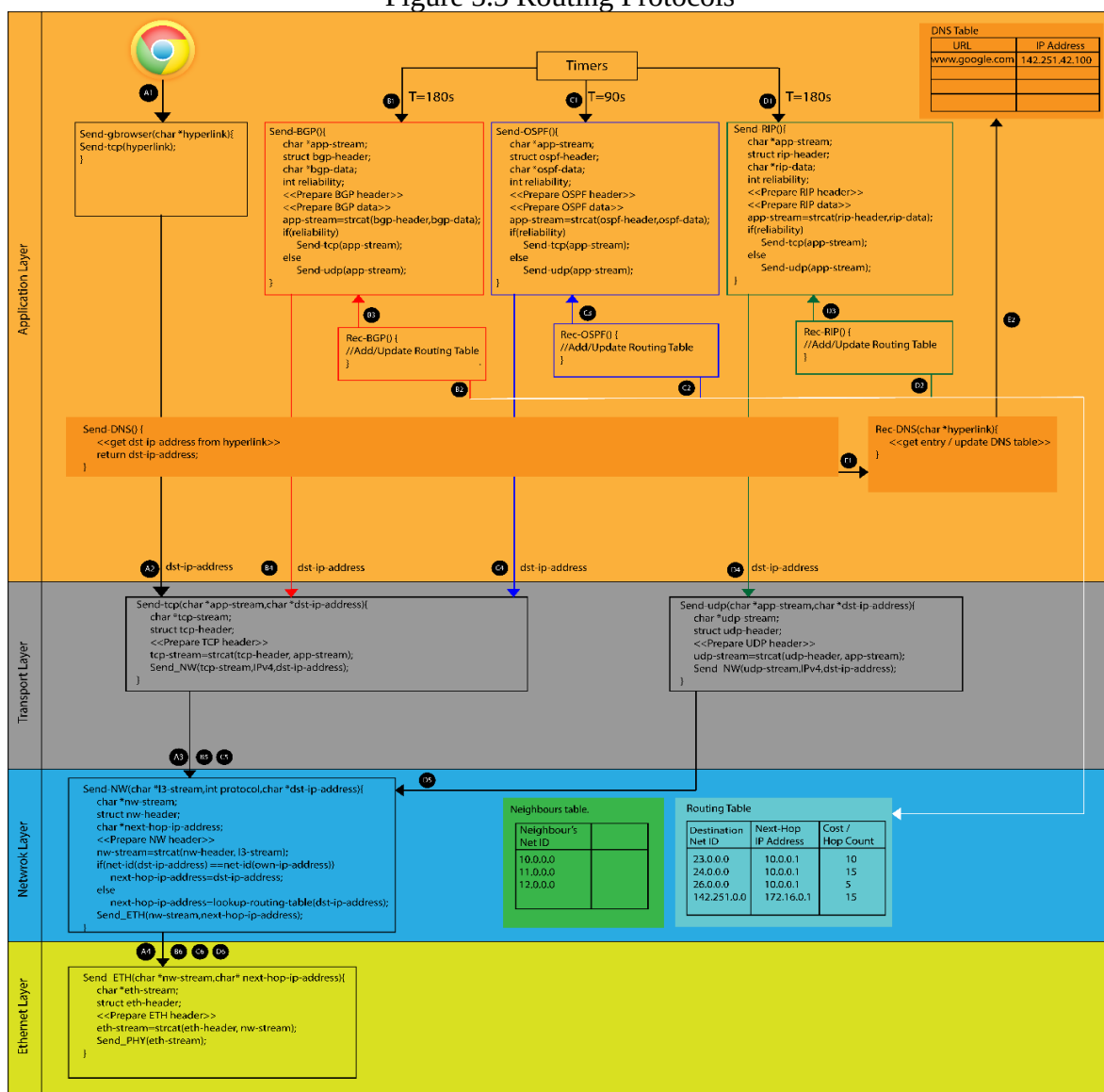
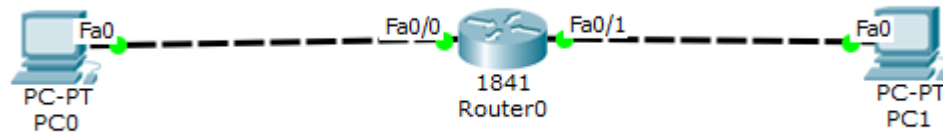
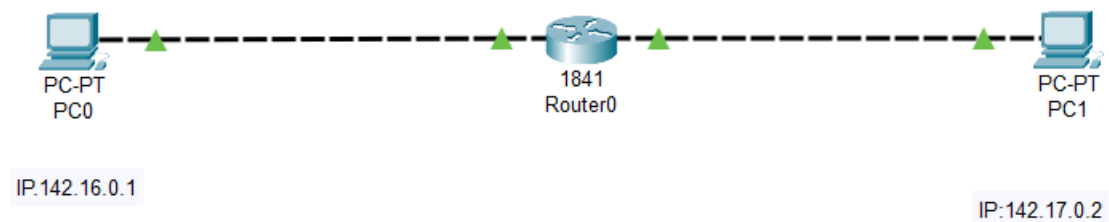


Figure 5.4 Demonstration of routing protocols in TCP/IP stack

## Exercise-1



Redraw above diagram which includes IP address and MAC address. Take IP address and MAC address as per your knowledge.



Insert Screenshots of ARP table of PC0 and PC1 before ping.

| ARP Table for PC0 |                  |           |
|-------------------|------------------|-----------|
| IP Address        | Hardware Address | Interface |
|                   |                  |           |
|                   |                  |           |
|                   |                  |           |
|                   |                  |           |
|                   |                  |           |
|                   |                  |           |
|                   |                  |           |
|                   |                  |           |
|                   |                  |           |

Ping from PC0 to PC1 and vice versa and insert snap of output here.

PC0 to PC1:

```
C:\>ping 142.17.0.2

Pinging 142.17.0.2 with 32 bytes of data:

Reply from 142.17.0.2: bytes=32 time<1ms TTL=127
Reply from 142.17.0.2: bytes=32 time<1ms TTL=127
Reply from 142.17.0.2: bytes=32 time<1ms TTL=127
Reply from 142.17.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 142.17.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC1 to PC0:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 142.16.0.1

Pinging 142.16.0.1 with 32 bytes of data:

Reply from 142.16.0.1: bytes=32 time=17ms TTL=127
Reply from 142.16.0.1: bytes=32 time<1ms TTL=127
Reply from 142.16.0.1: bytes=32 time<1ms TTL=127
Reply from 142.16.0.1: bytes=32 time<1ms TTL=127

Ping statistics for 142.16.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms
```

Insert Screen Shots of ARP table of PC0 and PC1 after ping

PC0:

| ARP Table for PC0 |                  |               |
|-------------------|------------------|---------------|
| IP Address        | Hardware Address | Interface     |
| 142.16.0.2        | 0060.2F78.9E01   | FastEthernet0 |

PC1:

ARP Table for PC1

| IP Address | Hardware Address | Interface     |
|------------|------------------|---------------|
| 142.17.0.1 | 0060.2F78.9E02   | FastEthernet0 |

Insert screenshots of ARP table and Routing Table at Router

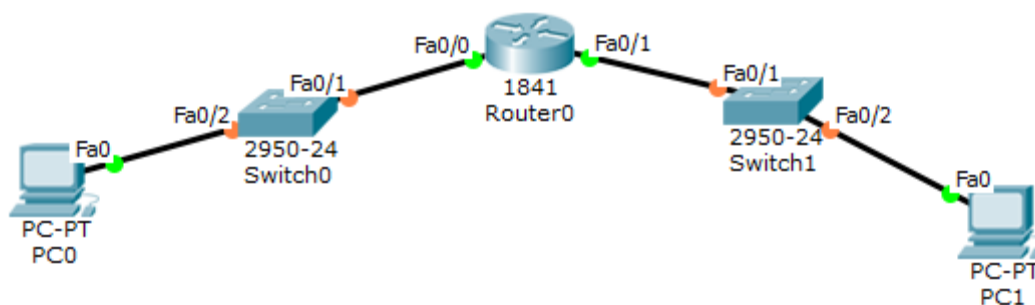
ARP Table for Router0

| IP Address | Hardware Address | Interface       |
|------------|------------------|-----------------|
| 142.16.0.1 | 0040.0BBA.C073   | FastEthernet0/0 |
| 142.16.0.2 | 0060.2F78.9E01   | FastEthernet0/0 |
| 142.17.0.1 | 0060.2F78.9E02   | FastEthernet0/1 |
| 142.17.0.2 | 000C.CF3E.3C71   | FastEthernet0/1 |

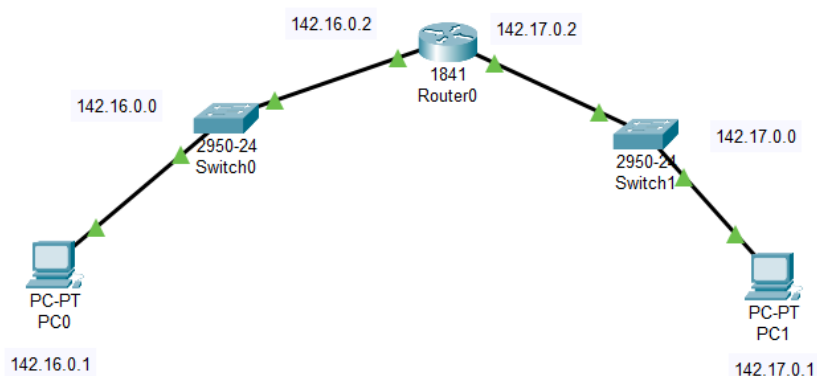
Routing Table:

Routing Table for Router0

| Type | Network       | Port            | Next Hop IP | Metric |
|------|---------------|-----------------|-------------|--------|
| C    | 142.16.0.0/16 | FastEthernet0/0 | ---         | 0/0    |
| C    | 142.17.0.0/16 | FastEthernet0/1 | ---         | 0/0    |

**Exercise-2**

Redraw above diagram which includes IP address and MAC address. Take IP address and MAC address as per your knowledge.



Insert Screenshots of ARP tables of all network components before ping.  
PC0:

| ARP Table for PC0 |                  |           |
|-------------------|------------------|-----------|
| IP Address        | Hardware Address | Interface |
|                   |                  |           |

PC1:

| ARP Table for PC1 |                  |           |
|-------------------|------------------|-----------|
| IP Address        | Hardware Address | Interface |
|                   |                  |           |

Ping from PC0 to PC1 and vice versa and insert snap of output here.

PC0 to PC1:

```
C:\>ping 142.17.0.1

Pinging 142.17.0.1 with 32 bytes of data:

Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.

Ping statistics for 142.17.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC1 to PC0:

```
C:\>ping 142.16.0.1

Pinging 142.16.0.1 with 32 bytes of data:

Reply from 155.17.0.2: Destination host unreachable.
Reply from 155.17.0.2: Destination host unreachable.
Request timed out.
Reply from 155.17.0.2: Destination host unreachable.

Ping statistics for 142.16.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Insert Screen Shots of ARP table of all network components after ping

PC0:

| ARP Table for PC0 |                  |               |
|-------------------|------------------|---------------|
| IP Address        | Hardware Address | Interface     |
| 142.12.0.2        | 000C.85D6.B401   | FastEthernet0 |

PC1:



ARP Table for PC1

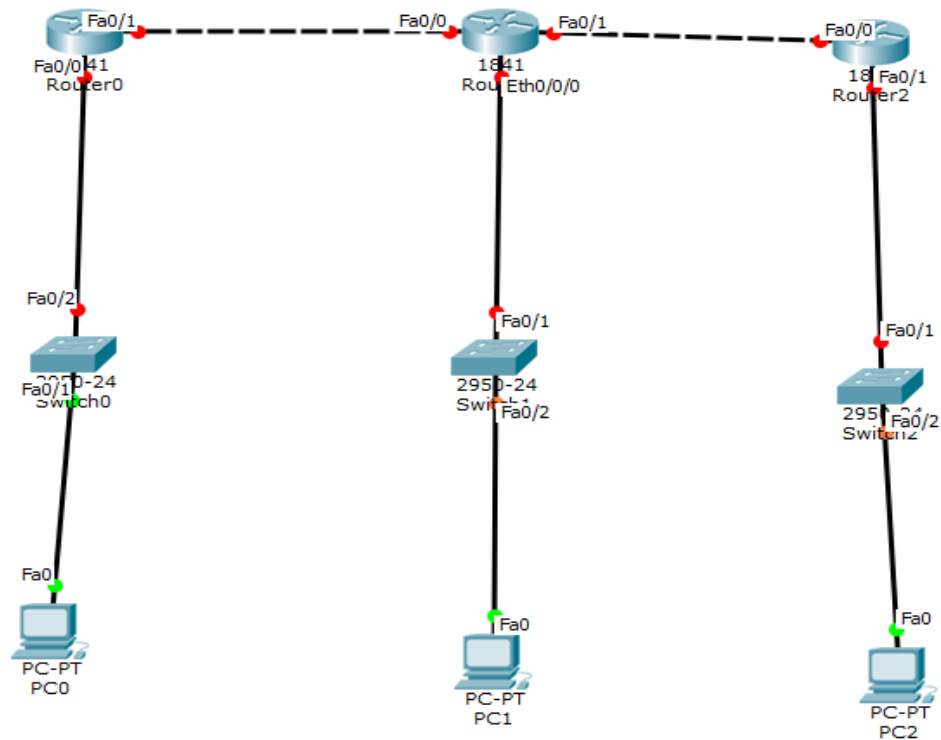
| IP Address | Hardware Address | Interface     |
|------------|------------------|---------------|
| 142.13.0.2 | 0002.17A3.7202   | FastEthernet0 |

Insert screenshots of Routing Table at Router

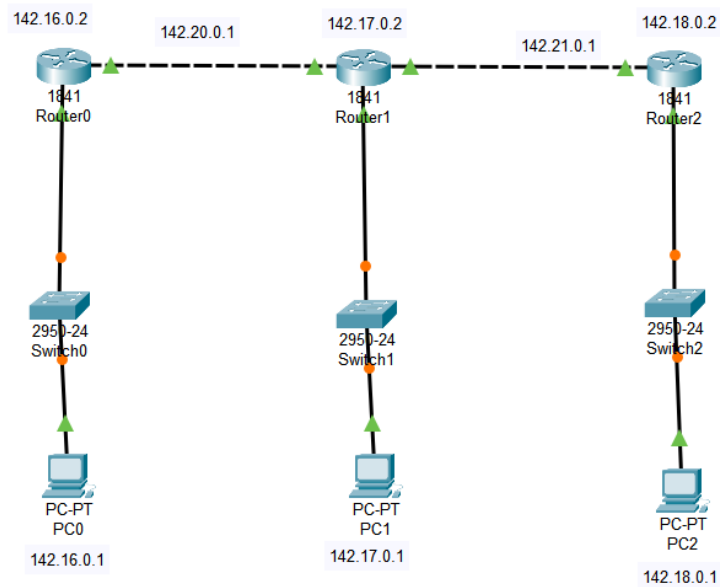
ARP Table for Router4

| IP Address | Hardware Address | Interface       |
|------------|------------------|-----------------|
| 142.12.0.1 | 0060.3E6D.28DE   | FastEthernet0/0 |
| 142.12.0.2 | 000C.85D6.B401   | FastEthernet0/0 |

### Exercise-3



Redraw the above diagram which includes the IP address and MAC address. Take the IP address and MAC address as per your knowledge.



Insert Screenshots of ARP tables of all network components before ping.

PC0:

| ARP Table for PC0 |                  |           |
|-------------------|------------------|-----------|
| IP Address        | Hardware Address | Interface |
|                   |                  |           |

PC1:

| ARP Table for PC1 |                  |           |
|-------------------|------------------|-----------|
| IP Address        | Hardware Address | Interface |
|                   |                  |           |

PC2:

| ARP Table for PC2 |                  |           |
|-------------------|------------------|-----------|
| IP Address        | Hardware Address | Interface |
|                   |                  |           |

Ping from PC0 , PC1 and PC2 and insert snap of output here.

PC0 to PC1:

```
C:\>ping 142.17.0.1

Pinging 142.17.0.1 with 32 bytes of data:

Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.

Ping statistics for 142.17.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC0 to PC2:

```
C:\>ping 142.18.0.1

Pinging 142.18.0.1 with 32 bytes of data:

Reply from 155.16.0.2: Destination host unreachable.
Request timed out.
Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.

Ping statistics for 142.18.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC1 to PC0:

```
C:\>ping 142.16.0.1

Pinging 142.16.0.1 with 32 bytes of data:

Reply from 155.17.0.2: Destination host unreachable.
Reply from 155.17.0.2: Destination host unreachable.
Request timed out.
Reply from 155.17.0.2: Destination host unreachable.

Ping statistics for 142.16.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC1 to PC2:

```
C:\>ping 142.18.0.1

Pinging 142.18.0.1 with 32 bytes of data:

Reply from 155.16.0.2: Destination host unreachable.
Request timed out.
Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.

Ping statistics for 142.18.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC2 to PC0:

```
C:\>ping 142.16.0.1

Pinging 142.16.0.1 with 32 bytes of data:

Reply from 155.17.0.2: Destination host unreachable.
Reply from 155.17.0.2: Destination host unreachable.
Request timed out.
Reply from 155.17.0.2: Destination host unreachable.

Ping statistics for 142.16.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC2 to PC1:

```
C:\>ping 142.17.0.1

Pinging 142.17.0.1 with 32 bytes of data:

Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.
Reply from 155.16.0.2: Destination host unreachable.

Ping statistics for 142.17.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Insert Screen Shots of ARP table of all network components after ping PC0:

ARP Table for PC0

| IP Address | Hardware Address | Interface     |
|------------|------------------|---------------|
| 142.12.0.2 | 000C.85D6.B401   | FastEthernet0 |

PC1:

ARP Table for PC1

| IP Address | Hardware Address | Interface     |
|------------|------------------|---------------|
| 142.13.0.2 | 0002.17A3.7202   | FastEthernet0 |

PC2:

ARP Table for PC2

| IP Address | Hardware Address | Interface     |
|------------|------------------|---------------|
| 142.14.0.2 | 00E0.F715.B901   | FastEthernet0 |

Router 0:

| ARP Table for Router4 |                  |                 |
|-----------------------|------------------|-----------------|
| IP Address            | Hardware Address | Interface       |
| 142.12.0.1            | 0060.3E6D.28DE   | FastEthernet0/0 |
| 142.12.0.2            | 000C.85D6.B401   | FastEthernet0/0 |
| 142.15.0.1            | 000C.85D6.B402   | FastEthernet0/1 |
| 142.15.0.2            | 0002.17A3.7201   | FastEthernet0/1 |

Router 1:

| ARP Table for Router5 |                  |                 |
|-----------------------|------------------|-----------------|
| IP Address            | Hardware Address | Interface       |
| 142.13.0.1            | 0001.63E0.9518   | FastEthernet0/1 |
| 142.13.0.2            | 0002.17A3.7202   | FastEthernet0/1 |
| 142.15.0.1            | 000C.85D6.B402   | FastEthernet0/0 |
| 142.15.0.2            | 0002.17A3.7201   | FastEthernet0/0 |
| 142.16.0.1            | 0002.4A04.EE98   | Ethernet0/0/0   |

Router 2:

| ARP Table for Router6 |                  |                 |
|-----------------------|------------------|-----------------|
| IP Address            | Hardware Address | Interface       |
| 142.14.0.1            | 0090.2BDC.7053   | FastEthernet0/0 |
| 142.14.0.2            | 00E0.F715.B901   | FastEthernet0/0 |
| 142.16.0.1            | 0002.4A04.EE98   | FastEthernet0/1 |
| 142.16.0.2            | 00E0.F715.B902   | FastEthernet0/1 |

Insert screenshots of Routing Tables of all Routers

Router 0:

Routing Table for Router4

| Type | Network       | Port            | Next Hop IP | Metric |
|------|---------------|-----------------|-------------|--------|
| C    | 142.12.0.0/16 | FastEthernet0/0 | ---         | 0/0    |
| S    | 142.13.0.0/16 | ---             | 142.15.0.2  | 1/0    |
| S    | 142.14.0.0/16 | ---             | 142.15.0.2  | 1/0    |
| C    | 142.15.0.0/16 | FastEthernet0/1 | ---         | 0/0    |

## Router 1:

Routing Table for Router5

| Type | Network       | Port            | Next Hop IP | Metric |
|------|---------------|-----------------|-------------|--------|
| S    | 142.12.0.0/16 | ---             | 142.15.0.1  | 1/0    |
| C    | 142.13.0.0/16 | FastEthernet0/1 | ---         | 0/0    |
| S    | 142.14.0.0/16 | ---             | 142.16.0.2  | 1/0    |
| C    | 142.15.0.0/16 | FastEthernet0/0 | ---         | 0/0    |
| C    | 142.16.0.0/16 | Ethernet0/0/0   | ---         | 0/0    |

## Router 2:

Routing Table for Router6

| Type | Network       | Port            | Next Hop IP | Metric |
|------|---------------|-----------------|-------------|--------|
| S    | 142.12.0.0/16 | ---             | 142.16.0.1  | 1/0    |
| S    | 142.13.0.0/16 | ---             | 142.16.0.1  | 1/0    |
| C    | 142.14.0.0/16 | FastEthernet0/0 | ---         | 0/0    |
| C    | 142.16.0.0/16 | FastEthernet0/1 | ---         | 0/0    |